

TRAFFIC LIGHT CONTROLLER

Verilog HDL Design and Vivado Simulation

1. Aim

Design a two-road traffic light controller using Verilog HDL.
Control North-South (NS) and East-West (EW) traffic lights.
Implement the controller using a Finite State Machine (FSM).
Verify the design using a Vivado simulation testbench.

2. Objectives

Create four traffic-light states.

Generate automatic transitions using a timer.

Use a sensor input to request an early change from EW Green.

Observe the outputs through Vivado behavioral simulation.

3. Block Diagram



4. FSM State Sequence

S0: NS Green, EW Red

S1: NS Yellow, EW Red

S2: NS Red, EW Green

S3: NS Red, EW Yellow

Sequence repeats continuously: $S0 \rightarrow S1 \rightarrow S2 \rightarrow S3 \rightarrow S0$

5. Output Encoding

State	NS	EW	Meaning
S0	3'b001	3'b100	NS Green / EW Red
S1	3'b010	3'b100	NS Yellow / EW Red
S2	3'b100	3'b001	NS Red / EW Green
S3	3'b100	3'b010	NS Red / EW Yellow

6. Verilog Design Code

```
module traffic_light_controller(  
    input clk, input reset, input sensor,  
    output reg [2:0] NS, output reg [2:0] EW  
);  
  
parameter S0=2'b00, S1=2'b01,  
          S2=2'b10, S3=2'b11;  
reg [1:0] state;  
reg [3:0] timer;  
  
always @(posedge clk or posedge reset)  
begin  
    if (reset) begin state <= S0; timer <= 0; end  
    else begin  
        timer <= timer + 1;  
        case (state)  
            S0: if (timer == 5) begin state <= S1; timer <= 0; end  
            S1: if (timer == 2) begin state <= S2; timer <= 0; end  
            S2: if (timer == 5 || sensor) begin state <= S3; timer <= 0; end  
            S3: if (timer == 2) begin state <= S0; timer <= 0; end  
        endcase  
    end  
end  
end
```

7. Output Logic

S0: NS = 3'b001; EW = 3'b100;

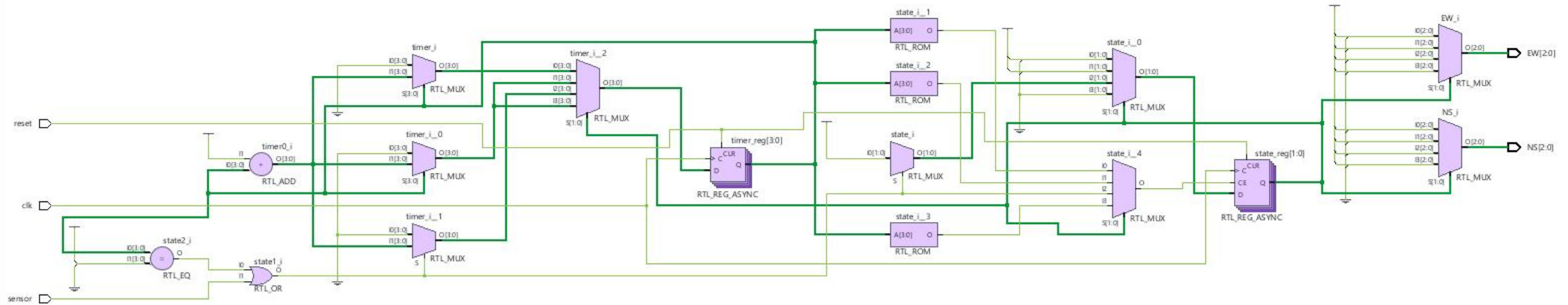
S1: NS = 3'b010; EW = 3'b100;

S2: NS = 3'b100; EW = 3'b001;

S3: NS = 3'b100; EW = 3'b010;

3'b001 = Green, 3'b010 = Yellow, 3'b100 = Red.

8. Vivado RTL Schematic



9. Testbench

```
`timescale 1ns/1ps
module traffic_tb;
reg clk, reset, sensor;
wire [2:0] NS, EW;

traffic_light_controller uut(
    .clk(clk), .reset(reset), .sensor(sensor),
    .NS(NS), .EW(EW)
);

always #5 clk = ~clk;

initial begin
    clk = 0; reset = 1; sensor = 0;
    #10 reset = 0;
    #80;
    sensor = 1;
    #20 sensor = 0;
    #100 $finish;
end
endmodule
```

10. Expected Simulation Output

NS: 001 → 010 → 100 → 100 → 001 ...

EW: 100 → 100 → 001 → 010 → 100 ...

The two directions alternate so both are not Green at the same time.

The sensor can cause the EW Green state to change to EW Yellow.

11. Result & Conclusion

The traffic light controller was successfully designed using Verilog HDL.

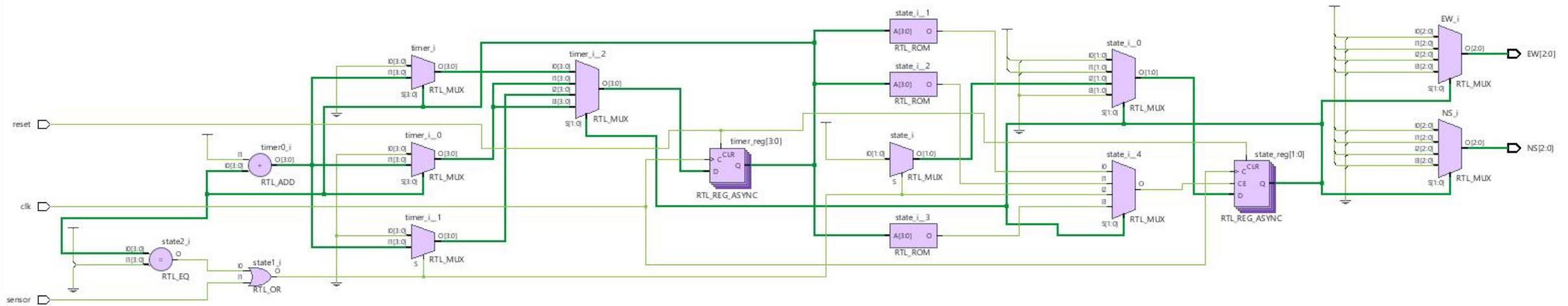
The FSM controls the sequence of NS and EW traffic lights.

The timer controls state transitions.

The design can be verified in Vivado using the testbench and waveform.

For physical FPGA implementation, the demo timer values can be replaced with a clock divider for real-time delays.

Vivado RTL Schematic Diagram



RTL schematic generated in Vivado from the Verilog traffic light controller.